

	Case Name: Broker Account Conversion	Sector	IT	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources Schedule with costs	
Submitted by	Matilde Casado and Margarida Antunes	Risk Analysis	Random simulation	
Date	2015		One of nine std. scenarios	
File Name	C2015-13 Broker Account Conversion	Project Control	User defined distributions	
			Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

A project involving requirements gathering, analysis, design, development, testing, and implementation to enhance data processing and system functionality.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	22	Serial/Parallel (SP)	19%
Planned Duration (PD)	117 days*	Activity Distribution (AD)	60%
Budget At Completion (BAC)	12.735 €	Length of Arcs (LA)	7%
Renewable Resources	7	Topological Float (TF)	51%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

Cost sensitivity				Time sensitivity			
	avg [%]	std dev [%]	skew [-]		avg [%]	std dev [%]	skew [-]
CRI-r	12.0	17.1	-2.6	CI	15.7	25.2	-2.0
CRI-rho	32.7	19.9	-0.3	SI	26.0	29.8	-0.7
CRI-tau	49.7	44.6	-0.2	SSI	7.2	17.6	-3.4
Resource sensitivity				CRI-r	9.4	17.1	-3.2
	avg [%]	std dev [%]	skew [-]	CRI-rho	30.3	22.4	-0.1
CRI-r	29.0	11.7	-1.7	CRI-tau	51.0	43.3	-0.2
CRI-rho	29.0	11.7	-1.7				
CRI-tau	29.0	11.7	-1.7				

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	505.1	-505.1	1	0.2	0.2
PV - SPI	497.1	-497.1	CPI	0.2	0.2
PV - SCI	486.7	-486.7	SPI	0.6	0.6
ED - 1	1987.2	-1987.2	SPI(t)	0.7	0.7
ED - SPI	497.1	-497.1	SCI	0.6	0.6
ED - SCI	487.5	-487.5	SCI(t)	0.7	0.7
ES - 1	71.2	-71.2	0.8 CPI + 0.2 SPI	0.4	0.4
ES - SPI(t)	28.8	-25.4	0.8 CPI + 0.2 SPI(t)	0.6	0.6
ES - SCI(t)	28.7	-25.4			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 4 tracking periods with a length of approximately 1 month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	1521.0	-1482.8	-114.5	1.2	0.7	0.7	0.9
std dev	1144.8	1327.6	128.2	0.1	0.3	0.4	0.1
final	2745.0	0	0	1.3	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	117 days	Real Duration	118 days	Late	0.85%
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EAC(t)	Real Accuracy			
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	136.3	1.0	3.9	-3.9
PV - SPI	136.8	0.5	4.0	-4.0
PV - SCI	137.3	0.5	4.1	-4.1
ED - 1	137.8	1.0	4.2	-4.2
ED - SPI	138.5	1.3	4.3	-4.3
ED - SCI	139.5	1.3	4.6	-4.6
ES - 1	140.5	1.3	4.8	-4.8
ES - SPI(t)	141.3	1.0	4.9	-4.9
ES - SCI(t)	141.8	0.5	5.0	-5.0

2.3.3.3. Cost forecasting

BAC	12.735 €	Real Cost	9 990.0 €	Under Budget	-21.6%
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EAC	Real Accuracy			
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	11214.0	1144.8	3.0	3.0
CPI	10554.5	847.2	4.3	4.3
SPI	16520.4	9257.6	7.4	-7.4
SPI(t)	71225.8	119173.4	114.8	-114.8
SCI	15344.3	8458.4	5.1	-5.1
SCI(t)	66108.4	110506.3	104.8	-104.8
0.8 CPI + 0.2 SPI	11106.7	1517.8	3.2	3.2
0.8 CPI + 0.2 SPI(t)	11330.3	2012.8	3.8	3.8